

DN: A Pre-Generation Based Interactive Narrative System with Full-Ready Multimodal Delivery

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Abstract

Interactive narrative systems increasingly combine language generation, state tracking, and scene-image synthesis, but this richer output scope can make player-facing latency a central bottleneck. We present DN, a pre-generation based interactive narrative system designed for *full-ready* multimodal delivery, where the next turn is counted as ready only when both the textual continuation and the corresponding scene image are available. On a 20-item formal benchmark, a simulated 60-second player reading interval reduces DN’s mean next-turn full-ready latency from 27.095 seconds to 5.299 seconds and its median latency from 24.571 seconds to 0.027 seconds, while preserving a strict success rate of 1.000. We further summarize released evidence that DN outperforms three image-generation baselines on judged visual quality, exceeds the current DOC fallback baseline on judged text coherence, and exhibits mechanism-level ablation patterns consistent with the proposed pre-generation design. Together, these results suggest that reading-time-aware pre-generation can make full-ready multimodal narrative delivery substantially more practical for interactive use.

1 Introduction

Interactive narrative systems must balance rich multimodal generation with low user-perceived latency. Unlike single-turn text or image generators, a playable narrative system must maintain story state, expose actionable options, advance the plot after a user decision, and deliver the next piece of content in a form that the player can immediately continue from. When the next turn contains both text and a scene image, the relevant efficiency question is therefore not merely how quickly

a model can produce an isolated image, but how quickly the system can make the complete next turn usable.

DN addresses this problem through a pre-generation based delivery mechanism. Instead of waiting until the user clicks the next action to begin all downstream work, DN uses the player’s reading interval to prepare likely next-turn content in the background. This design turns reading time into computation time and aims to reduce the latency that remains visible to the player. The present draft evaluates this mechanism across efficiency, visual quality, narrative/text quality, and component-level ablations, so that the paper can jointly address whether DN is fast enough for interaction and whether its delivered content is strong enough for narrative play.

The current efficiency results support three stage-level findings. First, strict full-ready delivery without dwell time remains expensive because the system must wait for both textual and visual readiness. Second, a realistic reading interval substantially changes the latency distribution: in the pregen60 setting, most samples are delivered nearly immediately after the next click. Third, comparisons against image-continuation references should be interpreted as scope-aware references rather than fully isomorphic system comparisons, because DN returns text plus image plus playable state while the baselines report image or workflow latency.

2 System Overview

DN is an interactive narrative generation system that combines structured story-state management, option generation, and scene-image delivery. At each turn, the system presents narrative text and choices to the player, re-

ceives the selected action, updates the story state, and prepares the next multimodal scene. The paper’s central efficiency hypothesis is that pre-generation can reduce user-visible delay by using the idle interval between content delivery and the player’s next action.

The pre-generation mechanism maintains candidate next-turn work in the background. During the player’s reading interval, DNprepares likely continuations and their associated scene images. When the player later selects an action, the system attempts to serve a strict full-ready result from the prepared content; if the requested content is not yet fully available, the measured path waits until both text and image are ready. This distinction is important for the evaluation: a response is not counted as successful merely because a placeholder or partial output exists.

3 Experimental Setup

We evaluate efficiency on a 20-item formal benchmark subset derived from the DN-quality benchmark. Each item defines a narrative theme and constraints that can be mapped either to DN’s full interactive pipeline or to image-generation baseline adapters. The evaluation uses the latest v15 composite20 artifacts as the source of truth and excludes earlier v14, fixed-v2, and trimmed-sample exploratory results from the main paper tables.

The primary comparison is not intended to claim that all systems are architecturally identical. Instead, it measures the latency of DN’s full-ready delivery and reports image-baseline latencies as reference points for the image-continuation subtask. This scope-aware framing is necessary because DNperforms story-state update and textual continuation in addition to scene-image delivery, whereas StoryDiffusion, SDM-v2, and IC-LoRA are evaluated as next-turn image or image-workflow references.

4 Efficiency Evaluation

Evaluation goal. We evaluate whether DNcan deliver next-turn content with practical latency while preserving consistent multimodal outputs. To keep one unified comparison surface, we report efficiency and consistency metrics in a single main table across the

same baseline set.

Protocol and metrics. The current efficiency values use the formal20 setup with strict full-ready timing for DNand next-turn image or workflow latency for image baselines. Consistency fields are kept in the same table schema but are marked as TODO until the Qwen3-based and DINO-v2-based runs are fully frozen under one script.

Main takeaways. Table 1 is the paper’s primary result table and replaces the previous split-table layout. The efficiency block already shows that DNremains competitive under a heavier output scope, while the consistency blocks reserve fixed slots for Qwen3 and DINO-v2 so later updates can be filled without changing table structure or interpretation logic.

5 Hyperparameter Configuration (Ongoing)

Status. Hyperparameter tuning is still in progress, so we provide a fixed reporting template instead of freezing incomplete numeric claims. This keeps the experimental section reproducible while making the remaining search space explicit.

How to update. Once tuning is complete, each TODO cell should be replaced directly from the corresponding run logs and config snapshots so the final camera-ready table remains a faithful record of the actually executed settings.

6 Visual Quality Evaluation

Evaluation goal. This section focuses on whether the delivered images are usable for narrative play, not only on speed. To avoid fragmented reporting, visual findings are now represented through the Image Consistency rows in Table 1.

Current evidence and reporting policy. We keep two image-consistency channels in the unified table schema: an LLM-based score (Qwen3) and a feature-model score (DINO-v2 cosine). Both are currently marked as TODO because the cross-system runs still need to be frozen under a single evaluator pipeline. Until that freeze is complete, we avoid introducing

Group	Metric	DN	StoryDiffusion	SDM-v2	IC-LoRA
Efficiency (lower is better, except Success Rate)					
	Mean Latency (s)	5.299	8.495	2.827	30.057
	P95 Latency (s)	27.150	8.900	2.831	30.059
	Success Rate	1.000	1.000	1.000	1.000
Image Consistency (higher is better)					
	LLM-based (Qwen3, 1–5)	TODO	TODO	TODO	TODO
	Model-based (DINO-v2 cosine)	TODO	TODO	TODO	TODO
Text Consistency (higher is better)					
	LLM-based (Qwen3, 1–5)	TODO	TODO	TODO	TODO
	Model-based (domain model)	N/A	N/A	N/A	N/A

Table 1: Main results under a unified protocol. Efficiency rows are populated from the latest formal20 summary. Consistency rows are placeholders pending unified evaluation artifacts: [PATH_TO_QWEN3_CONSISTENCY_RESULTS] and [PATH_TO_DINO_CONSISTENCY_RESULTS].

Module	Hyperparameter	Current Value	Search Range	Notes
Story planner	TODO	TODO	TODO	In progress
Pregeneration	TODO	TODO	TODO	In progress
Image generation	TODO	TODO	TODO	In progress
Consistency judge (Qwen3)	TODO	TODO	TODO	In progress
Consistency proxy (DINO-v2)	TODO	TODO	TODO	In progress

Table 2: Hyperparameter table template (ongoing). Pending artifact path: [PATH_TO_HYPERPARAM_LOGS].

additional standalone visual tables and keep all visual claims scoped to the same main-results structure.

Interpretation. This consolidation ensures that visual consistency is interpreted side by side with latency and text consistency, so later updates can be compared on a common baseline axis rather than across heterogeneous table layouts.

7 Narrative and Text Quality Evaluation

Evaluation goal. The text-side question is whether DN keeps coherent and stable narrative progression across turns. We therefore align text-quality reporting with the same unified table used for efficiency and visual consistency.

Current evidence and reporting policy. Text consistency is represented in Table 1 with two slots: Qwen3-based judged consistency (to be filled) and model-based consistency (currently N/A for these baselines because no shared domain-specific automatic scorer is frozen yet). This avoids mixing partially aligned subsets into separate headline tables

and keeps the comparison contract explicit.

Interpretation. By reporting text consistency in the same matrix as latency and image consistency, the paper can discuss trade-offs across modalities directly and update remaining TODO entries without changing the experimental narrative structure.

8 Ablation and Mechanism Analysis

Evaluation scope. The ablation study isolates whether key mechanisms contribute to both responsiveness and consistency. We compare the full system against variants that remove one mechanism at a time.

Interpretation. Table 3 keeps the ablation axis directly aligned with the main-results metrics so each mechanism can be discussed in terms of both efficiency and consistency. The currently filled rows already show the dominant contribution of pregeneration to latency reduction, while remaining TODO entries are reserved for the same evaluation contract once the council and read-wait reruns are finalized.

9 Conclusion

The current paper draft now supports a coherent scope-aware conclusion across efficiency, visual quality, and text-side quality. First, the efficiency results show that strict full-ready next-turn delivery remains expensive when no dwell time is available, but that pre-generation substantially changes the user-visible latency profile under a realistic reading

Variant	Mean Latency (s)	P95 Latency (s)	Image Consistency (Qwen3, 1–5)	Text Consistency (Qwen3, 1–5)
Full model	5.299	27.150	TODO	TODO
– pregen	27.095	46.196	TODO	TODO
– council	TODO	TODO	TODO	TODO
– readwait	TODO	TODO	TODO	TODO

Table 3: Ablation summary for mechanism contribution. The table isolates the effect of pre-generation, council planning, and read-wait strategy under aligned metrics. Pending artifact path: [PATH_TO_ABLATION_SUMMARY_CSV].

interval. On the latest v15 composite20 protocol, DN reduces mean full-ready latency from 27.095 seconds in the zero-dwell normal20 setting to 5.299 seconds in the pregen60 setting, while preserving a strict success rate of 1.000. This result establishes pre-generation as a practically meaningful systems mechanism rather than a cosmetic optimization.

Second, the quality sections indicate that faster delivery is not purchased at the cost of unusable outputs. On the released judged image-comparison summary, DN scores above StoryDiffusion, SDM-v2, and IC-LoRA, while its internal CLIP-I and DINOv2 image-pair proxies show moderate but non-trivial protagonist–scene continuity. On the text side, DN also performs strongly on the currently executable judged subset against the DOC fallback baseline, and the aligned GenAgents notes show that the remaining trade-off is primarily one of system orientation—structured planning and full-system stability versus lighter-weight persona-conditioned response—rather than simple coherence failure.

Third, the available mechanism analyses suggest that the present configuration of DN is not accidental. The main-ready evidence shows that reading-time-aware pre-generation materially improves full-ready delivery, while earlier controlled studies further suggest that pre-generation, council planning, and read-wait strategy each affect latency structure, stability, or the probability of returning a real scene instead of a weaker placeholder path. These auxiliary results are not all frozen under the same protocol as the main v15 composite20 table, but they remain useful for interpreting why the full system behaves as it does.

Overall, the current evidence supports the following paper-level claim: DN is a structured interactive narrative system that can

deliver full-ready multimodal next turns with stable success, and its pre-generation mechanism makes this heavier output scope substantially more compatible with interactive use. The remaining work is to further unify the protocol coverage of the auxiliary ablations, enlarge judged quality subsets, and continue reducing long-tail latency and theme-sensitive visual variance.

References

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- Yupeng Zhou, Daquan Zhou, Ming-Ming Cheng, Jiashi Feng, and Qibin Hou. 2024. *Storydiffusion: Consistent self-attention for long-range image and video generation*. *arXiv preprint arXiv:2405.01434*.

A Experiment Artifact Provenance

The efficiency tables in Section 4 use the latest v15 composite20 artifacts. Earlier exploratory runs, including v14, fixed-v2, and trimmed 16-sample variants, are retained only for internal analysis and are not used in the main paper numbers.

B Per-Sample Efficiency Details

The repository includes a per-item v15 composite20 table that records latency and mechanism signals for each benchmark item under both the zero-dwell normal20 setting and the simulated-read-time pregen60s20 setting. Table 5 summarizes the slowest cases from

Artifact	Role in Paper	Path
strict full-ready comparison	Main cross-system efficiency table	experiments/handoff_2026-04-30_main_experiments/01_main_tables/strict_fullready_main_comparison_2026-05-01_v6_composite20.csv
DN normal vs. pregen60	DN mechanism table	experiments/handoff_2026-04-30_main_experiments/01_main_tables/dn_fullready_normal_vs_pregen60_2026-05-01_v4_composite20.csv
v15 full-ready summary	Summary validation and mechanism signals	experiments/handoff_2026-04-30_main_experiments/01_main_tables/benchmark_v15_fullready_nextturn_20_summary_composite20_v1.json
per-sample v15 table	Appendix per-item inspection	experiments/benchmark/standard_runs/benchmark_v15_fullready_nextturn_20_table_composite20_v1.csv

Table 4: Source-of-truth artifacts for the efficiency evaluation. Paths are repository-relative.

each profile. This compact appendix view is intended to expose long-tail behavior without reproducing the full 40-row CSV inside the paper.

The appendix tail cases reinforce the main efficiency interpretation. Under `normal20`, the slowest items are dominated by rows whose cache-miss reason is `request_scene_generation_in_progress`, meaning that the second click arrives before the relevant scene generation path has fully completed. Under `pregen60s20`, most items become near-instant likely hits, and the remaining slow cases are concentrated in a small number of cache-miss or image-not-ready rows such as `DNQBV1_018`, `DNQBV1_014`, and `DNQBV1_009`. This is consistent with the aggregate result that pre-generation sharply improves the center of the latency distribution while leaving a smaller long tail that remains sensitive to scene readiness timing.

Profile	Benchmark ID	Theme	Full-ready (s)	Cache / readiness pattern	Drift flags
normal20	DNQBV1_012	深海信标失联	49.796	generation in progress	key+scene drift
normal20	DNQBV1_013	北极科考站怪声	46.006	generation in progress	key+scene drift
normal20	DNQBV1_011	书院禁书风波	32.181	generation in progress	key+scene drift
normal20	DNQBV1_019	维多利亚钟表匠	31.459	generation in progress	key+scene drift
normal20	DNQBV1_020	沙漠商队星图	28.806	generation in progress	key+scene drift
pregen60s20	DNQBV1_018	民国电报迷案	48.501	cache miss, waited for image	no drift
pregen60s20	DNQBV1_014	雨林部落预言	26.026	cache miss, waited for image	no drift
pregen60s20	DNQBV1_009	茶肆说书人	23.037	cache miss, waited for image	no drift
pregen60s20	DNQBV1_007	意识上传遗嘱	5.513	text ready, image not ready in cache	key+scene drift
pregen60s20	DNQBV1_008	全息偶像塌房记	2.551	text ready, image not ready in cache	key+scene drift

Table 5: Representative long-tail cases from the per-item v15 composite20 efficiency table. The rows are selected by highest `full_ready_elapsed_s` within each DN profile.